

B.Sc. Minor (Semester I) Mid Semester Examination, 2024-25
STAMN12: Practical

Time: 2 hours

Max. Marks: 30

*Note: Answer any **TWO** questions. All questions carry equal marks.*

Ques 1) The following numbers give the weight of 55 students in the class.

42	74	40	60	82	115	41	61	75	83	63
53	110	76	84	52	67	65	78	77	56	95
68	69	104	80	79	79	54	73	59	81	100
66	49	77	90	84	76	48	64	69	70	80
72	50	79	58	103	96	51	86	78	94	71

- Summarize the data in a grouped frequency distribution using the formula given by H. A. Sturges.
- Draw the histogram and frequency polygon of the given data. Also, obtain the arithmetic mean from the obtained frequency distribution.

Ques 2) Suppose the observations on X and Y are given as:

X	59	65	45	52	60	62	70	55	45	49
Y	75	70	55	65	60	69	80	65	59	61

where $N = 10$ students, and $Y =$ Marks in Maths, $X =$ Marks in Economics. Compute the least square regression equations of Y on X . If the student gets 61 marks in Economics, what would you estimate his marks in Maths to be? Also, give the interpretation of the coefficient of X .

Ques 3) Find the first four raw moments, central moments, coefficient of skewness and coefficient of Kurtosis for the data given in the table below and comment on the nature of the distribution. Also, find the corrected values of moments after Sheppard's corrections are applied.

Scores	Frequency
130-140	4
140-150	4
150-160	2
160-170	5
170-180	10
180-190	11

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